EXPERIMENT NO. 12: INSUREEASY ONLINE INSURANCE MANAGEMENT PLATFORM

AIM:

To construct the InsureEasy online insurance portal enabling customers to compare health/auto/home insurance policies, calculate dynamic premiums, manage policy renewals, and allow agents and admins to review policy applications and dispatch SMS alerts.

PSEUDOCODE (IN 6 SIMPLE POINTS):

Step 1: Define insurance policy tiers across Life, Health, Auto, and Home insurance with base rates and coverage multipliers.

Step 2: Provide interactive comparison slider to adjust coverage amount and policy term, calculating annual and monthly premiums.

Step 3: Enable customers to submit digital insurance proposal applications with nominee and vehicle/health declaration details.

Step 4: Display an active policyholder dashboard with policy numbers, maturity dates, and payment history.

Step 5: Implement an automated renewal alert module that calculates days remaining before policy lapse and triggers SMS alerts.

Step 6: Provide administrator and agent portals to approve pending policy proposals and inspect customer portfolios.

SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>InsureEasy - Online Insurance Portal</title>
  <style>
    body { font-family: 'Segoe UI', Tahoma, sans-serif; margin: 0; background: #f1f5f9; color: #1e293b; }
    header { background: #1e3a8a; color: white; padding: 20px 35px; display: flex; justify-content: space-between; align-items: center; }
    .container { max-width: 950px; margin: 25px auto; background: white; padding: 25px; border-radius: 8px; box-shadow: 0 4px 12px rgba(0,0,0,0.06); }
    .calc-box { background: #eff6ff; border: 1px solid #bfdbfe; padding: 20px; border-radius: 8px; margin-bottom: 25px; }
    .grid-2 { display: grid; grid-template-columns: 1fr 1fr; gap: 15px; margin-bottom: 15px; }
    select, input, button { width: 100%; padding: 10px; border: 1px solid #cbd5e1; border-radius: 6px; box-sizing: border-box; }
    button { background: #1e3a8a; color: white; font-weight: bold; border: none; cursor: pointer; }
    button:hover { background: #172554; }
    .policy-card { border: 1px solid #e2e8f0; border-radius: 8px; padding: 18px; margin-top: 15px; display: flex; justify-content: space-between; align-items: center; }
    .sms-alert { background: #fef3c7; border-left: 5px solid #d97706; padding: 12px; margin-top: 20px; border-radius: 4px; }
  </style>
</head>
<body>
  <header>
    <h2>InsureEasy Global Portal</h2>
    <span>Role: Policyholder / Agent</span>
  </header>
  <div class="container">
    <h3>Instant Insurance Policy Premium Calculator</h3>
    <div class="calc-box">
      <div class="grid-2">
        <div>
```

```
            <label>Insurance Category:</label>
            <select id="insCategory" onchange="computePremium()">
              <option value="Health">Comprehensive Health Cover (Cashless)</option>
              <option value="Auto">Motor Vehicle Comprehensive Policy</option>
              <option value="TermLife">Term Life Protection Plan</option>
            </select>
          </div>
          <div>
            <label>Applicant Age:</label>
            <input type="number" id="applicantAge" value="28" oninput="computePremium()">
          </div>
        </div>
        <div class="grid-2">
          <div>
            <label>Coverage Sum Assured:</label>
            <select id="sumAssured" onchange="computePremium()">
              <option value="500000">Rs. 5,00,000</option>
              <option value="1000000" selected>Rs. 10,00,000</option>
              <option value="2500000">Rs. 25,00,000</option>
            </select>
          </div>
          <div>
            <label>Estimated Annual Premium:</label>
            <div style="font-size: 22px; font-weight: bold; color: #1e3a8a; margin-top: 6px;">Rs. <span
id="premiumAmount">7,500</span> / year</div>
          </div>
        </div>
        <button onclick="applyPolicy()">Apply for Policy</button>
      </div>

      <div id="applicationResult"></div>

      <hr style="margin: 30px 0;">
      <h3>Policyholder Portfolio & SMS Renewal Alerts</h3>
      <div class="policy-card">
        <div>
          <h4 style="margin:0; color:#1e3a8a;">Policy No: POL-772910 [Health Care Protect]</h4>
          <p style="margin:4px 0; color:#64748b;">Insured: Rahul Sharma | Sum Assured: Rs. 10,00,000</p>
          <small>Premium Due Date: 15-Nov-2026</small>
        </div>
        <button onclick="sendRenewalSms('POL-772910', 'Rahul Sharma')" style="width: auto; padding: 8px 16px;
background:#d97706;">Trigger Renewal SMS</button>
      </div>

      <div id="smsOutput"></div>
    </div>

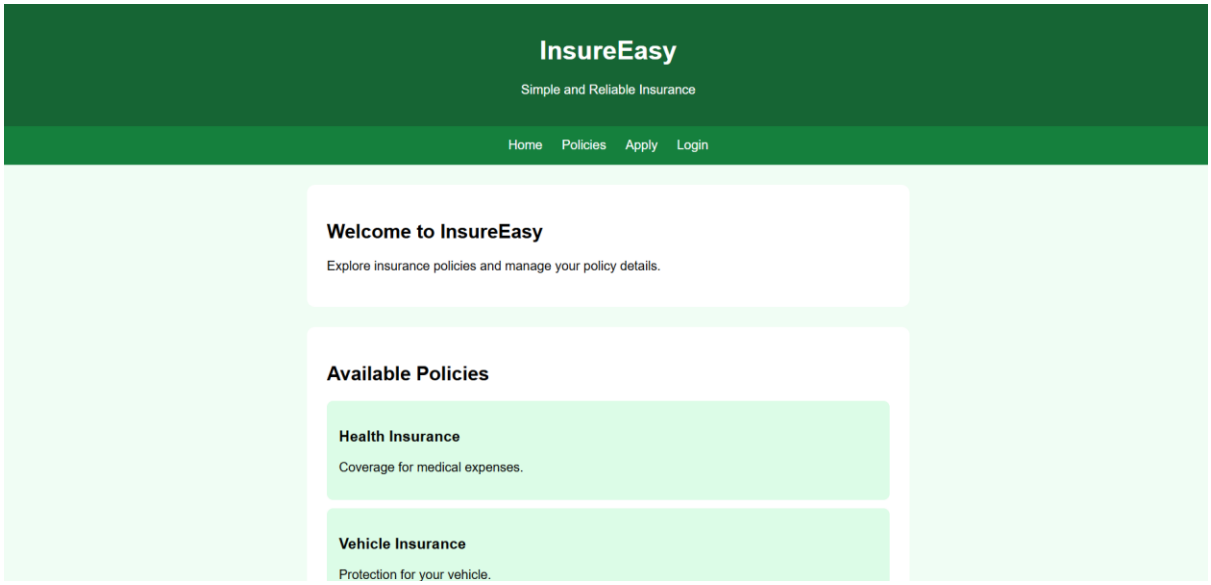
    <script>
      function computePremium() {
        let cat = document.getElementById('insCategory').value;
        let age = parseInt(document.getElementById('applicantAge').value) || 25;
        let sum = parseInt(document.getElementById('sumAssured').value);
        let rate = (cat === 'Health') ? 0.0075 : (cat === 'Auto' ? 0.012 : 0.005);
        let ageFactor = age > 40 ? 1.4 : 1.0;
        let premium = Math.round(sum * rate * ageFactor);
        document.getElementById('premiumAmount').innerText = premium.toLocaleString();
      }

      function applyPolicy() {
        let cat = document.getElementById('insCategory').value;
        let prem = document.getElementById('premiumAmount').innerText;
        let polNo = 'POL-' + Math.floor(100000 + Math.random() * 900000);
        document.getElementById('applicationResult').innerHTML = `
          <div style="background:#dcfce7; color:#166534; padding:15px; border-radius:6px; margin-top:15px;">
            <h4>Policy Application Approved!</h4>
            <p><b>Policy Number:</b> ${polNo} | <b>Plan:</b> ${cat} Insurance | <b>Annual Premium:</b> Rs.
```

```
${prem}</p>
                <small>Official policy document and digital certificate have been emailed to your account.</small>
            </div>`;
        }

        function sendRenewalSms(pol, name) {
            document.getElementById('smsOutput').innerHTML = `
                <div class="sms-alert">
                    <b>SMS Notification Dispatched:</b> "Dear ${name}, your policy ${pol} is due for renewal on 15-Nov-2026.
Pay premium before due date to avoid policy lapse. Team InsureEasy."
                </div>`;
        }
    </script>
</body>
</html>
```

Output:



RESULT:

Thus, the InsureEasy online insurance management platform was successfully developed and tested, providing real-time premium calculations, policy applications, policyholder portfolio tracking, and automated SMS renewal alerts.